Hugo Marquerie 14/12/2025

La inclusión en una localización es un morfismo de anillos

Proposición 1. *Sea R un ACU y $S \subset R$ una parte multiplicativa, entonces la aplicación*

$$\iota_S \colon R \longrightarrow S^{-1}R, \quad r \longmapsto \frac{r}{1}$$

es un morfismo de anillos con $\ker(\iota_S) = \{r \in R : \exists s \in S : sr = 0\}$.

***Demostración*:** Veamos que ι_S es un morfismo de anillos. Sean $r_1, r_2 \in R$, entonces

(i) $\iota_S(r_1 + r_2) = \dfrac{r_1 + r_2}{1} = \dfrac{r_1}{1} + \dfrac{r_2}{1} = \iota_S(r_1) + \iota_S(r_2)$.

(ii) $\iota_S(r_1 r_2) = \dfrac{r_1 r_2}{1} = \dfrac{r_1}{1} \cdot \dfrac{r_2}{1} = \iota_S(r_1) \cdot \iota_S(r_2)$.

(iii) $\iota_S(1) = \dfrac{1}{1} = 1_{S^{-1}R}$.

Por tanto, ι_S es un morfismo de anillos.

Sea $r \in \ker(\iota_S)$, entonces $\dfrac{r}{1} = \dfrac{0}{1}$ en $S^{-1}R$, luego $\exists s \in S : s(r \cdot 1 - 0 \cdot 1) = sr = 0$. ■

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.